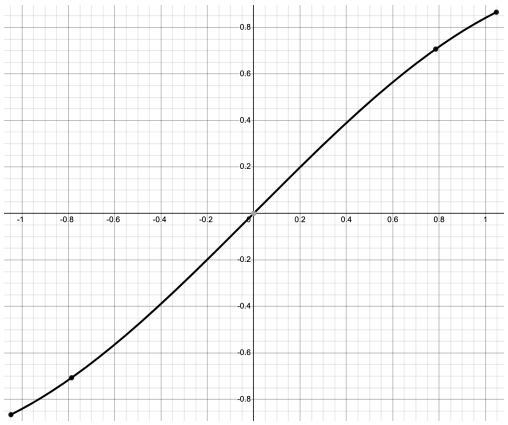
1. Graph the function $y = x^2$. Explain your technique.

For the graph of x^2, you likely plotted a few points and then assumed the behavior of the function on other points based on these observations. However, "proof by examples" is not a valid technique, as it can lead us astray. For example, suppose we tried to find the graph of $y = \sin(x)$ by plotting the points in the table below. From these points alone, we might also conclude that $\sin(x)$ is always increasing as $x \to \infty$, which is false.

x	$\sin(x)$
$-\frac{\pi}{3}$	$-\frac{\sqrt{3}}{2}$
$-\frac{\pi}{4}$	$-1/\sqrt{2}$
0	0
$\frac{\pi}{4}$	$1/\sqrt{2}$
$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$

(a) A few points on the graph of $y = \sin(x)$.



(b) The corresponding portion of the graph of $y = \sin(x)$.

2. So how do we *really* know that x^2 is always increasing as $x \to \infty$? That is, how do we know that there is not some humongous positive number x_0 out there after which x^2 begins decreasing? Discuss at your table for 1-2 minutes and jot down some thoughts.

First of all, before we can prove that "x^2 is increasing as $x \to \infty$", we need to state precisely what we mean by saying that a function $y = f(x)$ is "increasing". That is, we need a mathematical definition of "increasing", rather than an intuitive one. Intuitively, $y = f(x)$ is increasing if its graph goes upward as we move to the right. This is easy to write as a mathematical definition:

Definition. A function $y = f(x)$ is *(strictly) increasing* on an interval I if $f(x_2) > f(x_1)$ whenever $x_1, x_2 \in I$ and $x_2 > x_1$. Equivalently, $f(x_2) - f(x_1) > 0$ whenever $x_1, x_2 \in I$ and $x_2 - x_1 > 0$.

3. Write a similar definition of a *decreasing* function.

Notice that the definitions given are given in terms of inequalities. One can therefore use algebra to prove that the required inequality holds on a given interval.

4. Finish the following proof that that $y = x^2$ is increasing on the interval $(0, \infty)$: Suppose that $x_1, x_2 \in (0, \infty)$ and $x_2 - x_1 > 0$. Then

   $f(x_2) - f(x_1) =$

5. Prove that $y = x^2$ is decreasing on the interval $(-\infty, 0)$.

Throughout the semester, we will see a theme of introducing new ideas intuitively at first, then turning these ideas into precise mathematical definitions, and finally using these mathematical definitions to prove general results.